

Depth / k regularisation variants

Question: how much does the choice of prior/penalty matter?

FlexNN options: NONE, DEPTH_PENALTY, ELBO (KL vs linspace(3,1) prior), COST_AWARE_ELBO (linspace(3,1) - λ * normalised_depth_cost).

ProbReg options: NONE, K_PENALTY, ELBO (KL vs linspace(3,1) over k in [2..max]).

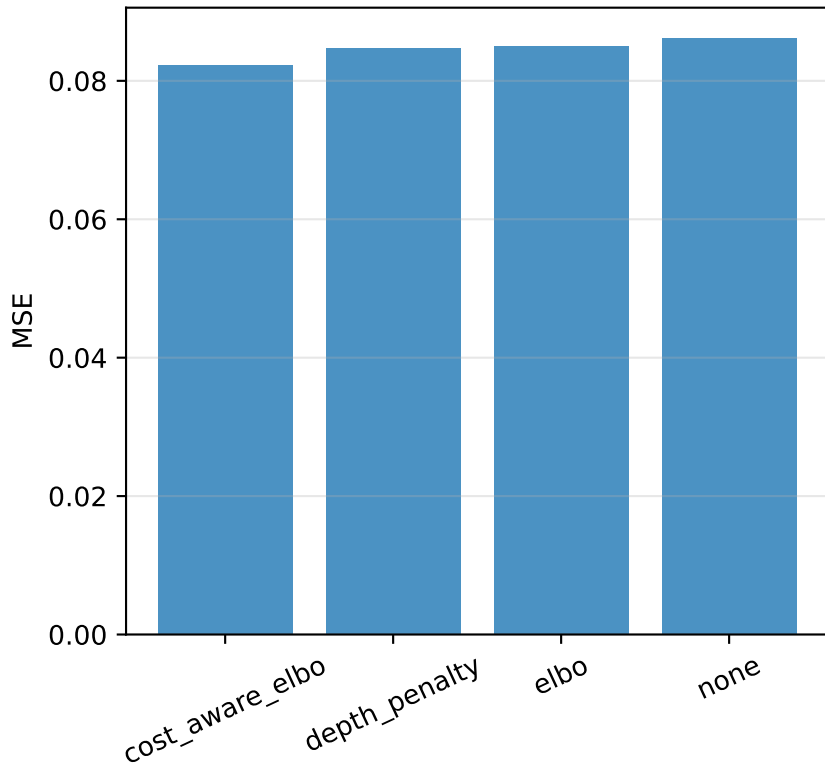
Expectation:

- DEPTH_PENALTY / ELBO should compress depth vs NONE.
- COST_AWARE_ELBO should further bias toward shallow on problems where the extra depth doesn't pay back in loss. On adaptive problems (piecewise, B3 two-phase) it should still pick deep layers where needed.
- ProbReg ELBO should pin k small (shown in `n_selection.pdf`).

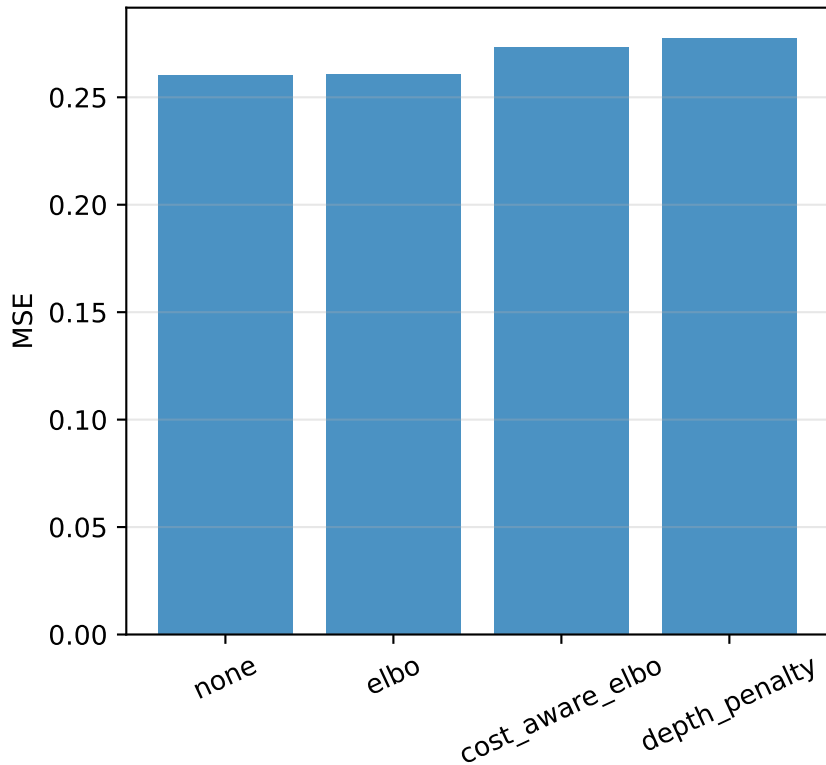
FlexNN: best MSE per depth regulariser

dataset	depth_reg	mse	mean_depth
b3_two_phase	cost_aware_elbo	0.082	1.790
b3_two_phase	depth_penalty	0.085	1.356
b3_two_phase	elbo	0.085	2.058
b3_two_phase	none	0.086	2.467
piecewise	cost_aware_elbo	0.273	1.774
piecewise	depth_penalty	0.278	1.016
piecewise	elbo	0.261	1.940
piecewise	none	0.260	1.051

b3_two_phase: min MSE per depth_reg



piecewise: min MSE per depth_reg



ProbReg: best MSE / NLL per regularisation (dyn-k cells only)

dataset	regularization	mse	nll
b1_gravitational	elbo	0.293	0.601
bimodal	elbo	2.328	1.847
heteroscedastic	elbo	1.623	1.378

Verdict — regularisation

Verdict: MET: COST_AWARE_ELBO is the MSE leader on B3 two-phase, and matches EL